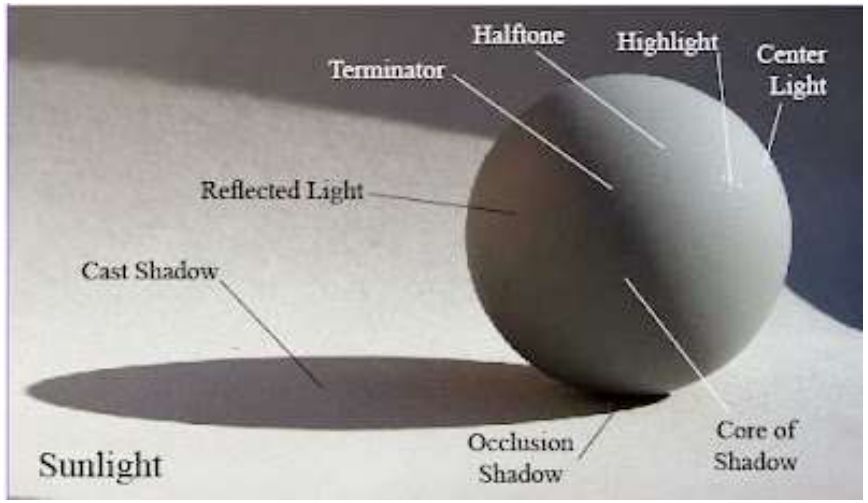
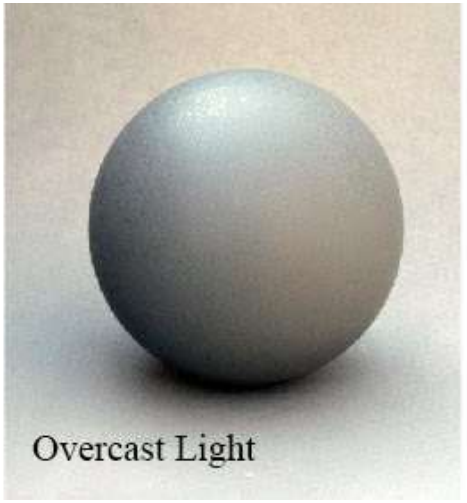
Observations from Color and Light

Color and Light

- Traditional painting lessons and their applications in digital illustration
- How do professional painters and illustrators capture color and light? Color and light are a concept that may be difficult to understand, as reality deceives the eye.
- Observations from *Color and Light: A Guide for the Realist Painter* by James Gurney

Overview of Basics

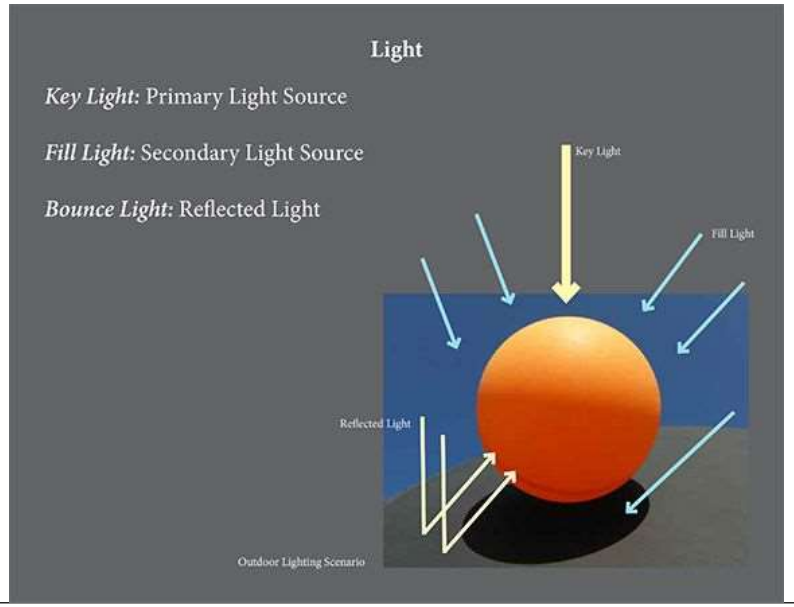




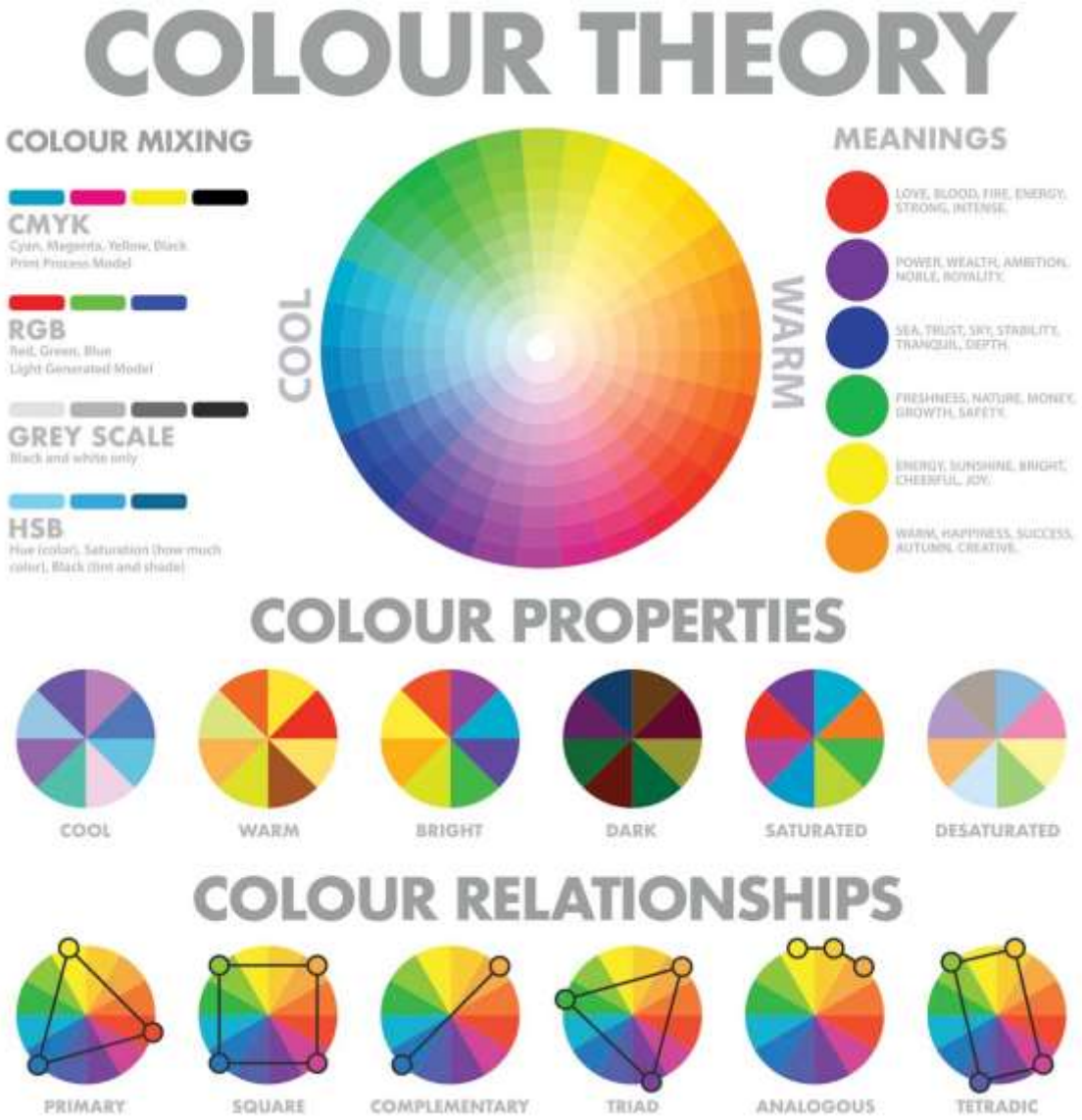
Sunlight creates a harsh division of light and shadow.

The *terminator* is where they are divided, often appearing darker than the rest of the shadow where it may be filled by reflected or fill light.

Overcast light diffuses light and shadow.



Overview of Basics



Themes and Keywords

Relativity – A color in comparison to those surrounding it. How we perceive color differs from its actual color in a photo.

Local color – The color of an object under white light

Color Constancy – A color depends on its environment. Mastering this will allow you to easily paint convincing scenic pictures

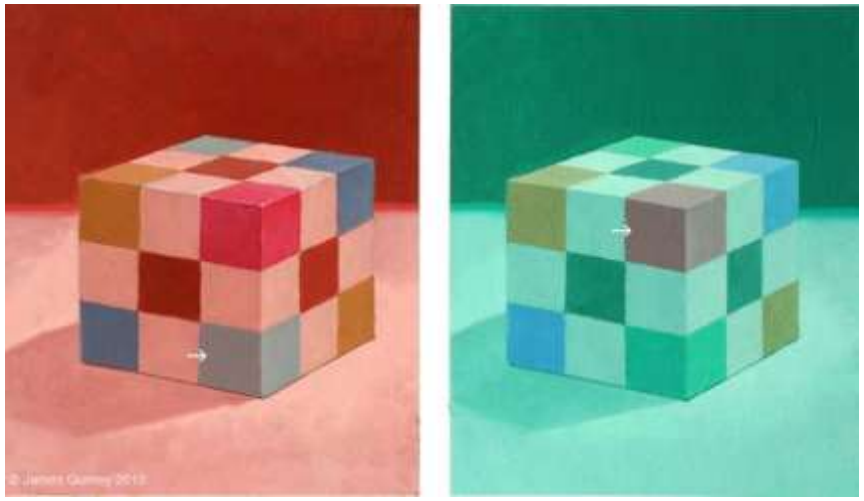
Light and Shadow – The form principle and its limitations, types of lighting and their uses. Choosing colors for shadows. Our eyes may have a hard time perceiving it, but shadows are everywhere.

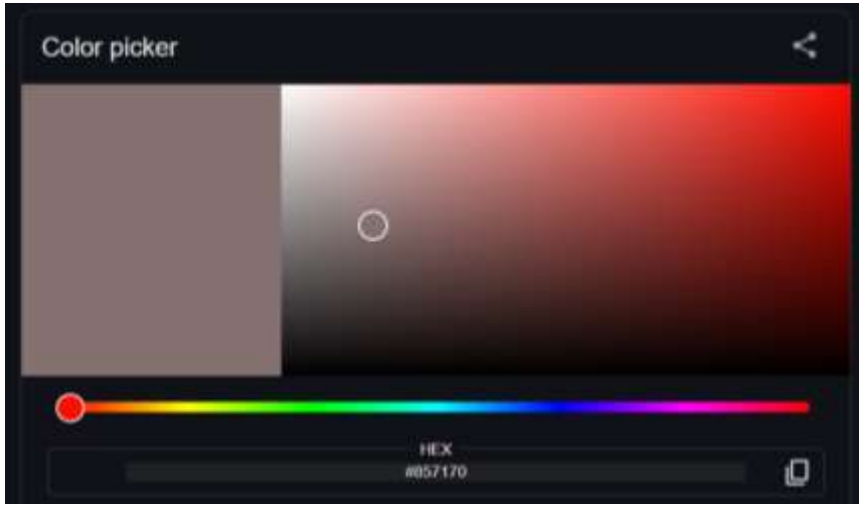
Texture and Detail – Texture is most visible in light and vague in shadow. Detail varies based on lighting conditions

Color – Beyond basic palettes, value and chroma, and the power of greys and neutrals.

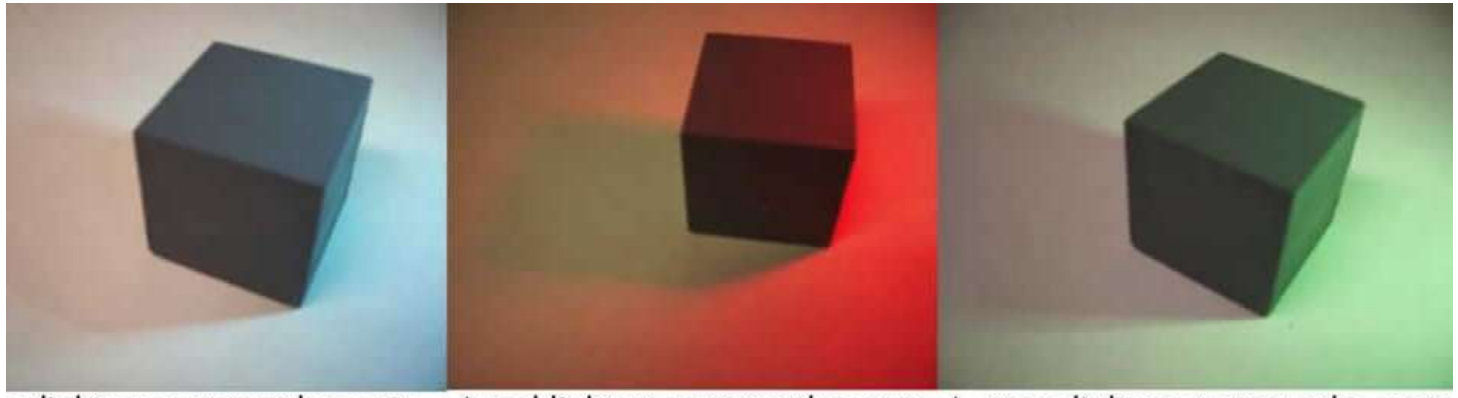
Color Constancy

- **Keyword *Relativity*** - a color's temperature, value, and whether it is “warm” or “cool” are affected by the colours around it rather than the colour itself.
- Colour constancy refers to an object's ability to appear the same colour under changing lighting conditions.
- Both highlighted spots in this James Gurney painting are painted with the same shade of red-grey, but they appear “blue” and “red” due to its **relative** shade to its surroundings. It is relatively cooler in the red light, and relatively warmer in the blue-green light (take note of the “white” cubes).
- Beginner painters tend to use colors literally, but it is rare that colors are what we may think they are.
- Color-picking from photos or paintings can help understand this concept.





Relativity and the Misconception of “Warm and Cool”



ıe light on a gray cube casts ·ange cast shadow. The ɔmatic cube appears to be

A red light on a gray cube casts a green cast shadow. The achromatic cube appears to have reddish tones.

A green light on a gray cube casts a reddish cast shadow. The achromatic cube appears to be green.

- Warm colors evoke feelings of sunsets, summer, fire, hot spices, blood. Cool colors evoke winter, night, sky, shadow, sleep and ice.
- Warm and Cool can refer to the **relative** tint of a color. For example, a cool yellow is a yellow that leans greener.
- Warm light produces **relatively** cool shadows.
- Cool light produces **relatively** warm shadows.
- Think of how a paint is mixed. Black paint is made by mixing the primary colors together, in other words mixing a color with its complimentary. Mixing blue paint with red and yellow paint darkens the color.

Shadow Color and Reflected Light

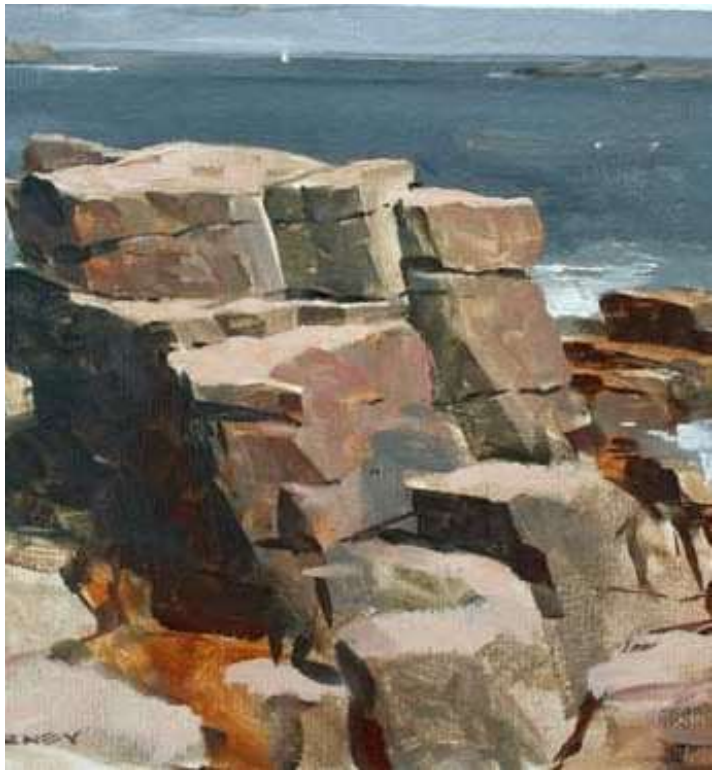
- On a bright sunny day, shadows are generally a **blue** color, filled by the blue sky.
- Surfaces that are **facing down** pick up **reflected light** from the surfaces below them, which are often warmer in tone as opposed to blue.
- In this James Gurney painting, the shadows are blue on the surfaces facing up or vertical, while the surfaces facing down such as the ceiling of the building are a warm color from the reflected light.



Texture and Detail in Lights

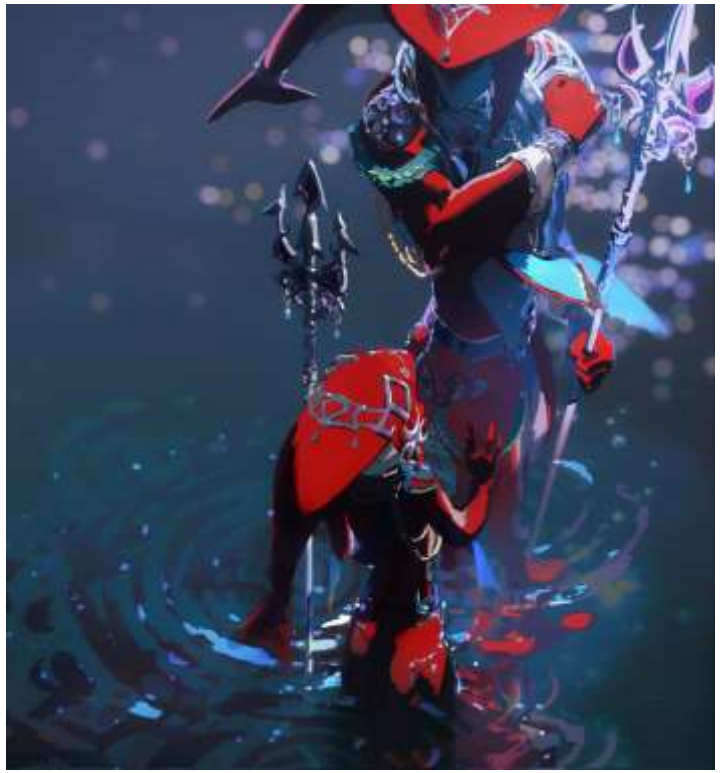
James Gurney





- Detail and texture are **most prominent under light** except for highlights. (E.g. sweater knits, reptile scales)
- The texture in shadow isn’t just a darker version of the texture in light. **Details fade beyond the terminator line.**
- Drawing texture in the shadows with equal prominence to the texture in light is laborious and less accurate.
- **Moonlit** scenes have **fewer details** than sunny ones as **fine details blend into larger shapes** due to our eyes' limited photoreception in the dark.
- Materials such as clouds and foliage let light pass through and may hot have a defined shadow and light.

Shadows and Colors





- In these illustrations by Redum (@REDUM4), the shadows are filled by their environments. Due to the harsh sunlight illuminating the lake, the green of the lake in the daylight painting reflects on the subject onto the shadow, giving it a relatively green tone.
- Distant objects fade due to atmospheric particles, appearing lower in contrast, detail, and tinted by the atmosphere. This effect is visible in the taller figure's faded shadows in the night painting, adding depth and reducing clutter.

Shadows and Colors

Top: Redum (REDUM4)
Bottom: KonYa666 (hxPTwPREsH6ll10)



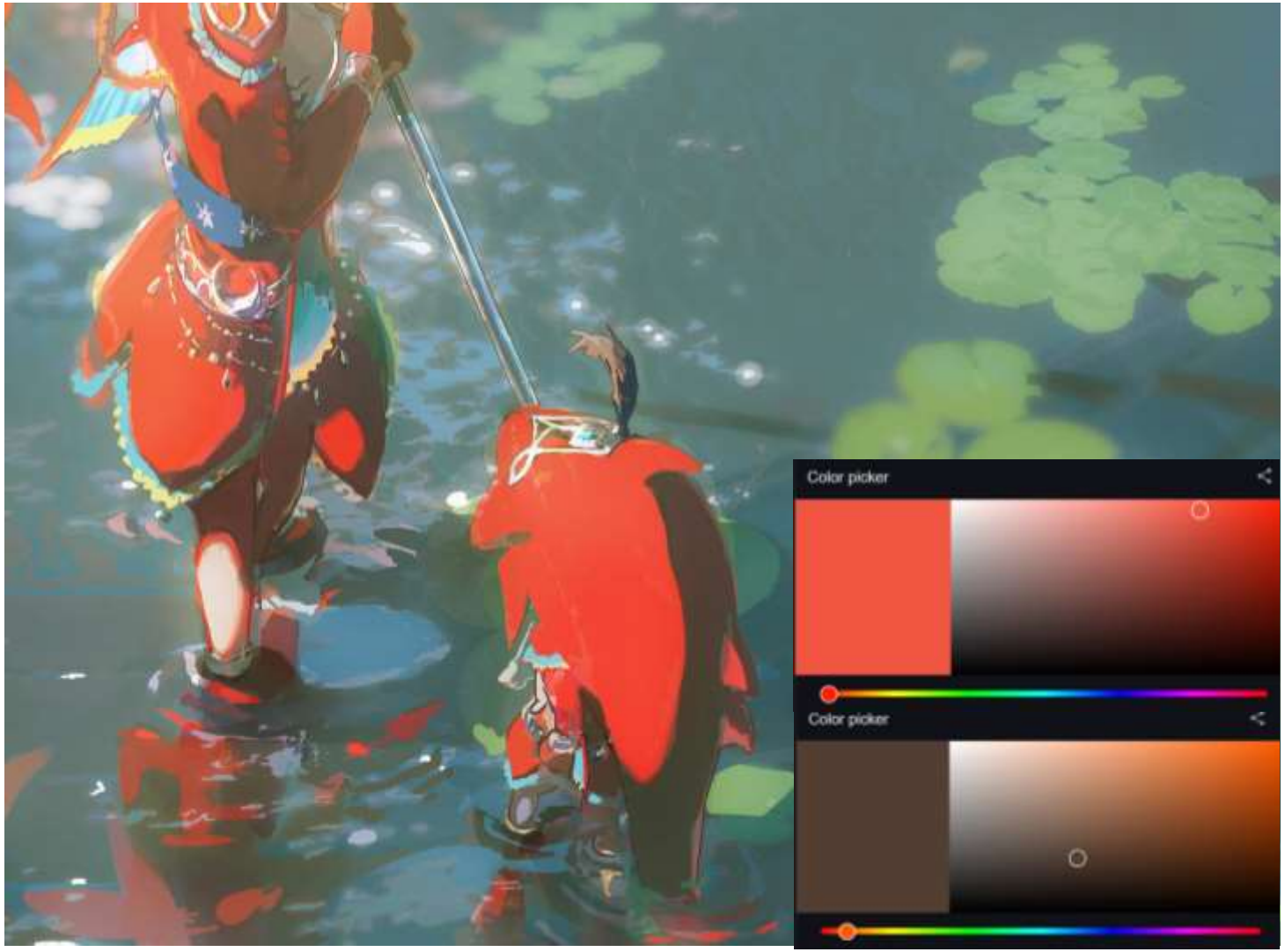




- Beginner painters tend to use shadow colors generally. Although shadows are blue on a sunny day, shadow color isn’t a universal shade.
- Shadow color is a sum of all the sources of reflected illumination. It picks up all the colors from its environment. It is blue on a sunny day due to its environment.

Observation of Actual Color

- The shadow on the red surfaces are green-tinted due to their surroundings, but their actual colors are dark, desaturated red-browns whose hue is tinted closer to the green side of the color spectrum.



Observations

- This James Gurney painting features contre-jour lighting with the subject against a bright light.
- The background is an overall shade of white, but the painting looks complete as it makes sense in the context of the lighting.
- Well-integrated lighting choices can make minimal backgrounds look finished.



Greys and Limited Palettes

- Greys are often associated with dullness and muddiness, and beginners may avoid it entirely. Greys and neutrals are actually indispensable for painting. ***More paintings fail because of too much intense color rather than too much grey.*** –Gurney
- Drabness or dullness is more often caused by poor value organization than by the colors themselves.
- Greys provide a setting for bright color accents. The eye is attracted to chromatic (saturation) contrast, so important areas can be highlighted and emphasized harmoniously with a high-chroma color against greys and neutrals.



Greys and Limited Palettes

- Left - Ahrii (ahriii7): Relatively warm grey accents in the shadows highlight the high-chroma blues and harmonize the painting.
- Right - 教主某狐仙 (kitune1132): Teal-tinted grey shadows compliment the high-chroma red.





Greys and Limited Palettes

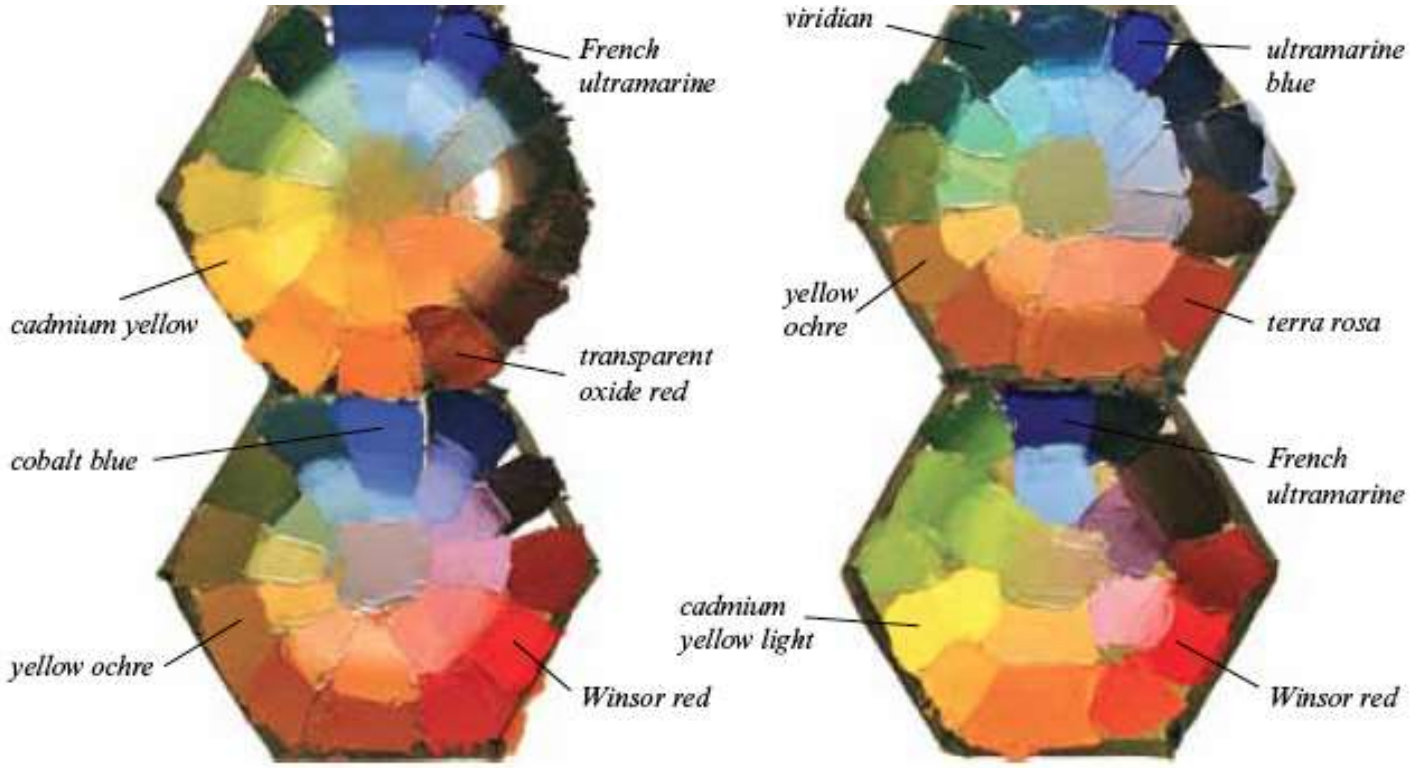
More colors don’t make a better color scheme. Limited palettes, a small selection of pigments, often result in a painting with a more unified or harmonious effect.

Old masters painted with limited color palettes as there weren’t as many paints as now.

It’s often a good idea to pick **one high-chroma color** and combine it with two **weaker colors** from across the spectrum.

(In traditional painting) Limited palettes break color-mixing habits. Mixing a color from scratch if you don’t have a color called "grass green" often leads to a more accurate green.

(In traditional painting) 4-5 colors are enough to paint most of nature.



Greys and Limited Palettes





Left -- 淅洵 (sPpQT4wRD3xa1KK)

This painting contains many high-chroma colors, but greys can be found throughout the shadows to balance them out.

Right – Ahrii (ahriii7)

This painting appears vivid but its color palette is limited. The colors stand out due to the greys and neutrals that take up most of the painting.

Some Interesting Lighting Choices

Illuminated foreground example, James Gurney



Rim lighting example, Kuro Games



- Illuminated Foreground – Casting the middle distance in shadow
- Hidden Light Sources – Light from a source that is obscured from view.
- Rim Lighting – Light from behind. (Note: Not just a white line around the form. Wider planes that face to the lit area will capture wider light.)
- Half Shadow – Top half in light, bottom half in shadow. Common in digital illustration, creates drama, depth and interest.